

Supplemental Material for Edge of Many-Body Quantum Chaos in Quantum Reservoir Computing

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I. DETAILED CHARACTERIZATION OF MANY-BODY QUANTUM CHAOS

In the main text, we characterize many-body quantum chaos by analyzing both the energy-level statistics and the spectral form factor (SFF). Here, we supplement those results with additional numerical data to provide a more comprehensive discussion.

A. Level statistics

The statistics of adjacent energy-level spacings provide a clear diagnostic of quantum chaos within the framework of random matrix theory (RMT). In particular, an integrable system exhibits a Poissonian distribution, whereas a quantum-chaotic system follows Wigner–Dyson (WD) statistics. The WD class splits further into three classes according to the fundamental symmetries of the Hamiltonian: Gaussian orthogonal ensemble (GOE), Gaussian unitary ensemble (GUE), and Gaussian symplectic ensemble (GSE), each labeled by the Dyson index $\beta = 1, 2$, and 4 . Their distributions of the level spacing ratio $P(r)$ are given by

$$P_{\text{Poi}}(r) = \frac{2}{(1+r)^2}, \quad P_{\text{WD}}^\beta(r) = \frac{2}{Z_\beta} \frac{(r+r^2)^\beta}{(1+r+r^2)^{1+3\beta/2}}, \quad (\text{S1})$$

with normalization constants $Z_1 = 8/27$, $Z_2 = 4\pi/(81\sqrt{3})$, and $Z_4 = 4\pi/(72\sqrt{3})$ [1]. The corresponding mean values are $\langle r \rangle_{\text{Poi}} = 0.3862$, $\langle r \rangle_{\text{GOE}} = 0.5307$, $\langle r \rangle_{\text{GUE}} = 0.5996$, and $\langle r \rangle_{\text{GSE}} = 0.6744$.

In the SYK₄ limit for the Hamiltonian in Eq. (1) of the main text, the level statistics adhere to GUE, as demonstrated by the agreement between the GUE prediction and the numerical data in Fig. 2(a) of the main text. This symmetry class arises because, for any finite N , particle-hole symmetry (PHS) is generally broken in the employed Hamiltonian. To see the effect of the symmetry, we restore PHS by incorporating the correction term

$$\mathcal{H}^{\text{PHS}} = \frac{1}{2} \sum_{i,j,k,l=1}^N J_{ijkl} (\delta_{ik} c_j^\dagger c_l - \delta_{il} c_j^\dagger c_k - \delta_{jk} c_i^\dagger c_l + \delta_{jl} c_i^\dagger c_k), \quad (\text{S2})$$

where $\delta_{i,j}$ is the Kronecker delta function [2, 3]. Figure S1(a) displays the distribution $P(r)$ in the half-filling sector $N_p = N/2$ for the SYK₄ model with the correction term, i.e., $\mathcal{H} + \mathcal{H}^{\text{PHS}}$ at $(J_4, \kappa_2) = (1, 0)$. It now follows GOE statistics, different from the GUE behavior in Fig. 2(a) of the main text,

reflecting the recovery of PHS. Naturally, $P(r)$ for the SYK₂ model follows Poisson statistics regardless of the PHS correction term. We also present the distribution for a system size $N = 7$ in Fig. S1(b), which reintroduces GUE behavior since PHS cannot be preserved for odd system sizes. Likewise, the inclusion of the SYK₂ term breaks PHS; even before the system reaches the integrable regime with Poisson distribution, $P(r)$ exhibits a gradual shift from GUE to GOE as the coupling strength κ_2/J_4 increases [4]. Since our primary aim is to explore the impact of many-body quantum chaos on information processing, symmetry-related discussion specific to the SYK models is out of the current scope. We thus utilized the Hamiltonian in Eq. (1) of the main text, which breaks PHS by construction.

As illustrated in Fig. 2(b) of the main text, continuously interpolating between the SYK₄ and SYK₂ models drives the system from a many-body quantum chaotic phase into an integrable regime. To elucidate this behavior in greater detail, Fig. S2(a) presents the full distribution $P(r)$ for several intermediate values of the coupling ratio κ_2/J_4 , extending the results presented in Fig. 2(a) of the main text. As κ_2/J_4 increases, $P(r)$ evolves smoothly from the WD (GUE) statistics toward the Poisson distribution. In Fig. S2(b), we plot the mean ratio $\langle r \rangle$ as a function of κ_2/J_4 for various system sizes at $N_p = \lfloor N/2 \rfloor$. The chaotic-integrable transition is consistently observed, although a precise determination of the boundary would require even larger system sizes [5]. For the

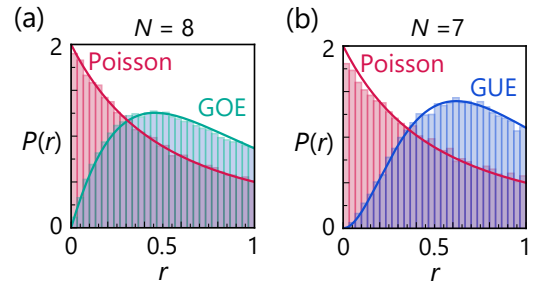


FIG. S1. (a), (b) Distribution of level spacing ratio $P(r)$ for the Hamiltonian including the PHS preserving term: $\mathcal{H} + \mathcal{H}^{\text{PHS}}$. Panel (a) corresponds to a system size of $N = 8$, while panel (b) uses $N = 7$. In both cases, the particle number sector is fixed at $N_p = \lfloor N/2 \rfloor$. Theoretical predictions for Poisson (red), GOE (green), and GUE (blue) statistics are shown as curves. The histograms present $P(r)$ for the SYK₂ model (red) and the SYK₄ model [green for (a) and blue for (b)]. Data are averaged over 2000 independent realizations in both panels.

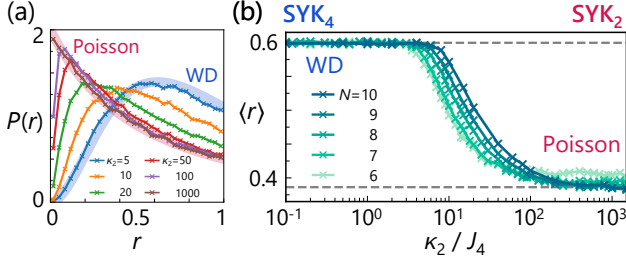


FIG. S2. (a) Distribution $P(r)$ for the Hamiltonian in Eq. (1) of the main text, with a fixed system size of $N = 8$. Curves are shown for several values of the coupling κ_2 at $J_4 = 1$. The background blue and red curves indicate WD (GUE) and Poisson predictions. Data are averaged over 2000 realizations. (b) Level spacing ratio $\langle r \rangle$ as a function of the coupling strength κ_2/J_4 for various system sizes. The particle number sector is fixed to $N_p = \lfloor N/2 \rfloor$. The dashed gray lines mark the reference value for WD (GUE) and Poisson statistics. Averages are taken over 2000 realizations for $N = 6, 7, 8, 1000$ for $N = 9$, and 500 for $N = 10$.

purposes of this work, we provisionally define the boundary of the quantum chaotic regime by the coupling strength at which $\langle r \rangle$ first deviates from its RMT value. This reference point is used to discuss the parametric edge of many-body quantum chaos in Figs. 4(a) and 4(b) of the main text.

B. Spectral form factor

The SFF $K(t)$ is a temporal domain probe used to capture the energy spectrum correlation across different energy scales. It is defined as the Fourier transform of the two level correlation function:

$$K(t) = \frac{1}{\mathcal{N}^2} \sum_{m,n} e^{i(E_m - E_n)t}, \quad (\text{S3})$$

where the sum runs over the eigenvalues with the particle number N_p , and $\mathcal{N} = \binom{N}{N_p}$ denotes the dimension of the corresponding Hilbert space. As discussed in the main text, the

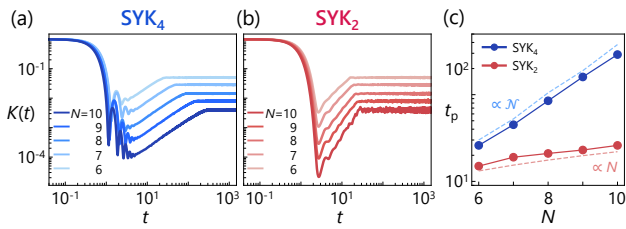


FIG. S3. (a), (b) The SFF for the SYK₄ model and the SYK₂ model, shown for various system sizes. Data are averaged over 20000 realizations for $N = 6, 7, 8, 10000$ for $N = 9$, and 2500 for $N = 10$. (c) The plateau time t_p of the SFF as a function of the system size N . The blue and red dashed lines indicate reference scaling behaviors, $\mathcal{N} = \binom{N}{N_p}$ and N , respectively.

SFF exhibits universal characteristics in the quantum chaotic systems: at early times, it decreases due to the phase interference, while at sufficiently late times, it shows a linear ramp before the saturation to a plateau value [6–8]. In Figs. S3(a) and S3(b), we present the SFF for the SYK₄ and SYK₂ models, respectively, for various system sizes. The ramp-plateau behavior is clearly demonstrated especially for the former system, confirming its many-body quantum chaotic nature. In contrast, the SFF for the latter model shows the dip and plateau over a much narrower time window. Indeed, Fig. S3(c) shows that the plateau time t_p scales with the system size as $t_p \sim \mathcal{N}$ for the SYK₄ system and as $t_p \sim N$ for the SYK₂ system [8, 9]. This contrast further highlights the difference between the quantum chaotic and integrable models. We note that the Thouless time in the SYK₄ model is known to scale as $t_{\text{Th}} \sim \sqrt{\mathcal{N}}$ [9], but the available sizes here are insufficient to reliably extract it.

When interpolating between the two models, the overall energy scale varies with the coupling strength. To compare QRC performance on an equal footing, we normalize the Hamiltonian by its spectral norm, $\mathcal{H} \rightarrow \mathcal{H}/\|\mathcal{H}\|$. This normalization leaves the level spacing ratio invariant, as well the chaotic nature of the system. However, it introduces a uniform prefactor in the temporal domain, affecting the SFF [Eq. (S3)] and the Δt_{in} dependency of QRC performance. Figure S4 displays the SFF for the interpolated model after the rescaling. In the quantum chaotic regime ($\kappa_2 = 0.1, 1$), the typical ramp-plateau profiles emerge, while in the integrable regime ($\kappa_2 = 20, 100$), the SFF dips and plateaus at much earlier times. Apart from the overall time-scale shift introduced by the Hamiltonian rescaling, these behaviors mirror those of the SYK₄ and SYK₂ models in Fig. 2(b) of the main text.

In our QRC performance benchmarks in the parametric domain (Fig. 4 of the main text), we selected the input time intervals $\Delta t_{\text{in}} = 50$ and $\Delta t_{\text{in}} = 1$; the corresponding times are marked by the dashed lines in Fig. S4. For the quantum chaotic regime, the longer Δt_{in} falls within the RMT ramping window, while the shorter one precedes the dip. Similarly for the integrable models, these two times effectively sandwich the dip. These observations validate both the Hamiltonian

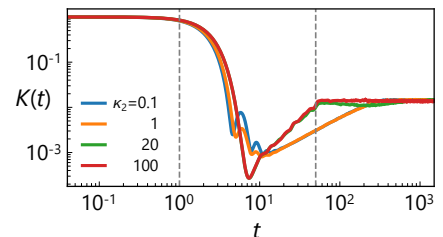


FIG. S4. The SFF for the rescaled Hamiltonian $\mathcal{H} \rightarrow \mathcal{H}/\|\mathcal{H}\|$ at several coupling strength κ_2 , with a fixed system size of $N = 8$ and $J_4 = 1$. The dashed lines signify $t = 1$ and 50, in relation to the input interval Δt_{in} used in to evaluate QRC performance. The results are averaged over 20000 realizations.

normalization procedure and our choice of Δt_{in} for probing distinct temporal regimes relevant to QRC. It is worth noting that precisely characterizing chaos in the interpolated model via the SFF typically requires a more sophisticated procedure called unfolding [10]. Unfolding recasts the energy spectrum such that it will exhibit a constant level spacing; it is not an operation directly applied to the Hamiltonian itself. We do not employ this procedure because our primary motivation is to characterize QRC performance by focusing on the properties of the original, physically implemented Hamiltonian.

II. DYNAMICS OF QUANTUM RESERVOIR SYSTEM

The QRC protocol consists of signal encoding, reservoir dynamics, read-out, and post-processing (see Fig. 1 of the main text). We briefly overview this protocol here to provide deeper insight into its dynamics. First, for a sequential input $\{u^{(k)}\}$, the k -th input $u^{(k)}$ is encoded into the N -site quantum reservoir system by replacing the quantum state at site 1. The system then evolves unitarily under the Hamiltonian for a duration Δt_{in} until the next input. During this evolution, the single-site particle number $\langle c_i^\dagger c_i \rangle$ is measured at V subintervals. The NV resulting read-out signals, along with an additional constant term, form the $(NV + 1)$ -dimensional state vector $\mathbf{x}^{(k)}$. The final output $y^{(k)}$ is obtained through a linear transformation of this vector with the trainable weight W_{out} .

Figures S5(a) and S5(b) illustrate the dynamics of read-out signals for the SYK₄ and SYK₂ models, respectively, for input intervals of $\Delta t_{\text{in}} = 1$ and $\Delta t_{\text{in}} = 10$. The dashed lines indicate when new inputs are provided. Markers show $\langle c_i^\dagger c_i \rangle$ at times $v\Delta t_{\text{in}}/V$ ($v = 0, \dots, V$), with the filled marker specifically de-

noting site 1, i.e., the input site. The dynamics at the input site change abruptly after an input, showing a larger variation range than other sites; hence, its vertical axis is displayed separately.

With a short input interval ($\Delta t_{\text{in}} = 1$), each new input arrives before the previous one has fully influenced the system. Thus, the resulting dynamics is relatively simple, as shown in the upper panels of both Figs. S5(a) and S5(b). In contrast, system-dependent characteristics emerge when the input interval is longer ($\Delta t_{\text{in}} = 10$). In the SYK₄ model [lower panel of Fig. S5(a)], all readout signals tend toward quasi-stationary values, whereas in the SYK₂ model [lower panel of Fig. S5(b)], the signals exhibit continuous oscillations. Notably, although both dynamics appear sufficiently rich, their QRC performance differs significantly, as demonstrated in Figs. 3(c) and 3(d) of the main text. Because the output $y^{(k)}$ is computed from these readout signals, this performance gap must originate from the difference in dynamics. However, the underlying reason for this performance gap is not immediately obvious from the raw signals in Fig. S5.

These Δt_{in} -dependent behaviors raise interest in the relationship between QRC performance and scrambling time—the timescale over which information delocalizes [11]. Since scrambling relates to the propagation of encoded input information, its connection to memory performance is naturally anticipated [12]. Accordingly, the observed Δt_{in} dependence of STM performance [upper panels of Figs. 3(c) and 3(d) of the main text] could plausibly reflect the underlying scrambling timescale. However, the pronounced enhancement in nonlinear processing performance [lower panel of Fig. 3(c) of the main text] cannot be explained by scrambling alone; scrambling itself is primarily associated with memory. Moreover, the convergence of QRC performance toward that of the Haar QRC model when Δt_{in} exceeds the Thouless time underscores a fundamental connection to RMT. These observations collectively suggest that the Thouless time plays the more decisive role in determining overall QRC performance. Nevertheless, the detailed relationship between scrambling time, Thouless time, and the temporal edge of many-body quantum chaos warrants further investigation.

III. CONVERGENCE PROPERTY

We here examine echo state property (ESP), a fundamental characteristic of RC systems. ESP dictates that the reservoir state should asymptotically converge when driven by the same input sequence, irrespective of its initial conditions. In other words, after sufficiently many input injections, any dependence on the initial state should vanish. In QRC paradigms, ESP is typically assessed by measuring the distance between two quantum states, $\rho_A^{(k)}$ and $\rho_B^{(k)}$. These states evolve from different initial states, $\rho_A^{(k=0)}$ and $\rho_B^{(k=0)}$, but under an identical input sequence. ESP is considered satisfied when this distance approaches 0. Under purely unitary dynamics, the distance between these states, evaluated using the Frobenius norm, re-

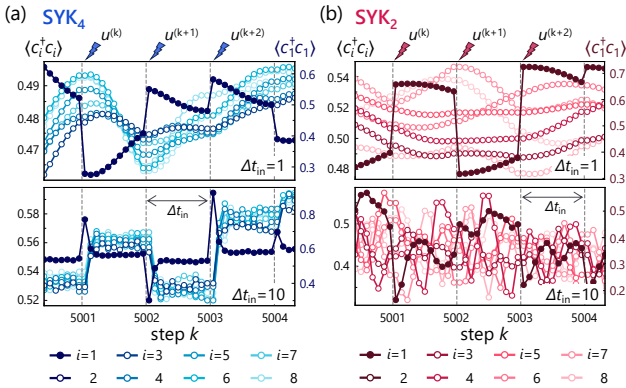


FIG. S5. (a), (b) Dynamics of the read-out signals $\langle c_i^\dagger c_i \rangle$ for the SYK₄ model and the SYK₂ model calculated for one typical realization. The upper panels correspond to the input interval $\Delta t_{\text{in}} = 1$, while the lower panels use $\Delta t_{\text{in}} = 10$. The vertical dashed lines indicate the input-encoding events $\rho^{(k-1)}(\Delta t_{\text{in}}) \rightarrow \rho^{(k)}(0)$. Data markers denote the values of $\langle c_i^\dagger c_i \rangle$ at times $v\Delta t_{\text{in}}/V$ where $0 \leq v \leq V$. The filled symbol corresponds to the value at the input site (scale on the right vertical axis), and the open symbols show values at the other sites (scale on the left vertical axis).

mains constant over time: $\|\rho_A^{(k)}(t) - \rho_B^{(k)}(t)\| = \|\rho_A^{(k)}(t') - \rho_B^{(k)}(t')\|$. Thus, the specific time t within a unitary evolution step is irrelevant. However, the distance does change upon the nonunitary input injection process, which involves the partial trace over site 1 and the reinitialization with the input state ρ_{in} . Consequently, to evaluate ESP, we monitor the state distance as a function of the input step k . As the system traces out the previous state and injects a new, identical input state for both trajectories, the distance between $\rho_A^{(k)}$ and $\rho_B^{(k)}$ is expected to decrease with k . Yet, the precise convergence behavior depends on the state before the input and the time evolution process $e^{-iH\Delta t_{\text{in}}}$. For instance, if the input interval is close to zero ($\Delta t_{\text{in}} \rightarrow 0$), the newly added input state is traced out almost immediately, and the distance remains largely unchanged. Conversely, a larger Δt_{in} yields more intricate dynamics, where the partial trace operation can lead to a more significant reduction in distance, the details of which depend on the Hamiltonian.

Figures S6(a) and S6(b) display the distance $\|\rho_A^{(k)} - \rho_B^{(k)}\|$ as a function of the input step k for various input intervals Δt_{in} in the SYK₄ and SYK₂ models, respectively. The initial states $\rho_{A,B}^{(0)}$ are randomly sampled from the half-filling sector. To accommodate values spanning several orders of magnitude, both the mean and median distances over 500 pairs of initial states are presented. For all models investigated, our results strongly indicate that the update rule [Eq. (2) of the main text] is contractive for random initial conditions and inputs: $\|\rho_A^{(k+1)} - \rho_B^{(k+1)}\| < \|\rho_A^{(k)} - \rho_B^{(k)}\|$. This observation suggests that ESP is satisfied for these systems—at least asymptotically as $k \rightarrow \infty$.

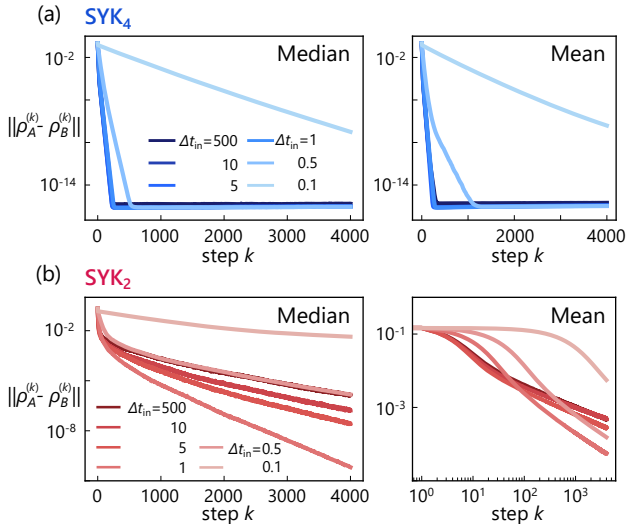


FIG. S6. Evolution of the distances $\|\rho_A^{(k)} - \rho_B^{(k)}\|$ as a function of the step k for (a) the SYK₄ model and (b) the SYK₂ model. The left panels show the median distance (semilog scale), and the right panels show the mean distance [semilog for (a) and log-log for (b)], calculated from 500 independent pairs of initial states $\rho_A^{(0)}$ and $\rho_B^{(0)}$. Colors indicate the input interval Δt_{in} .

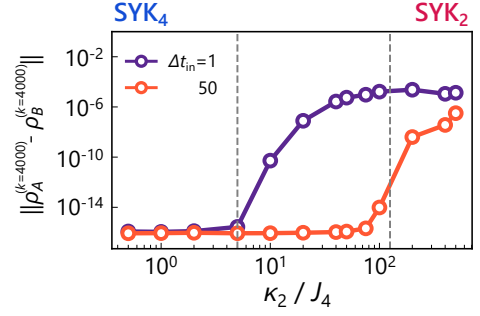


FIG. S7. Median distance $\|\rho_A^{(k)} - \rho_B^{(k)}\|$ at $k = 4000$ versus the coupling strength κ_2 / J_4 . Medians are calculated from 500 independent pairs of random initial states $\rho_A^{(0)}$ and $\rho_B^{(0)}$, and are presented for two input intervals: $\Delta t_{\text{in}} = 1$ (purple) and $\Delta t_{\text{in}} = 50$ (orange). The vertical lines indicate the onset of WD (left) and Poisson (right) statistics.

The convergence speeds of the SYK₄ and SYK₂ models differ significantly. In the SYK₄ model, the distance metric consistently decays exponentially toward 0, but the decay rate is sensitive to the input interval Δt_{in} . For smaller intervals ($\Delta t_{\text{in}} = 0.1, 0.5$), both the mean and median distances converge relatively slowly due to less effective erasure of past states upon the partial trace. Conversely, larger Δt_{in} values result in much faster convergence, with the distance behaving similarly across these larger input intervals. In stark contrast, the SYK₂ model exhibits considerably slower distance decay. While its median distance also decreases exponentially, the mean distance decays only polynomially and remains relatively high. This behavior reflects the integrable nature of the SYK₂ model, wherein numerous internal conserved quantities can impede convergence. These conserved quantities also generate large statistical fluctuations, which explains the disparity between the mean and median distances [Fig. S6(b)] and the large standard deviation in the performance metric (the SYK₂-dominated regime in Fig. 4 of the main text).

Figure S7 displays the median distance $\|\rho_A^{(k)} - \rho_B^{(k)}\|$ for the interpolated model as a function of the coupling strength κ_2 / J_4 . The distance is evaluated at the final step considered ($k = 4000$) and is presented for two input intervals: $\Delta t_{\text{in}} = 1$ and $\Delta t_{\text{in}} = 50$. Larger final distances indicate slower convergence, whereas smaller ones suggest rapid decay. Consistent with Figs. S6(a) and S6(b), the distance approaches 0 for small κ_2 / J_4 and remains relatively large in the strong κ_2 / J_4 regime for both input intervals. However, the behavior differs between the two input intervals at intermediate κ_2 / J_4 values. For $\Delta t_{\text{in}} = 1$, the distance deviates from 0 at relatively small κ_2 / J_4 , near the onset of WD statistics. In contrast, with $\Delta t_{\text{in}} = 50$, it stays close to 0 until κ_2 / J_4 approaches the boundary of the Poisson regime. This comparison indicates that convergence becomes harder to achieve as the system nears the integrable regime, which requires a large Δt_{in} to generate the necessary complex dynamics.

We note that Ref. [13] analyzes QRC performance in spin-based quantum reservoirs, attributing the poor performance

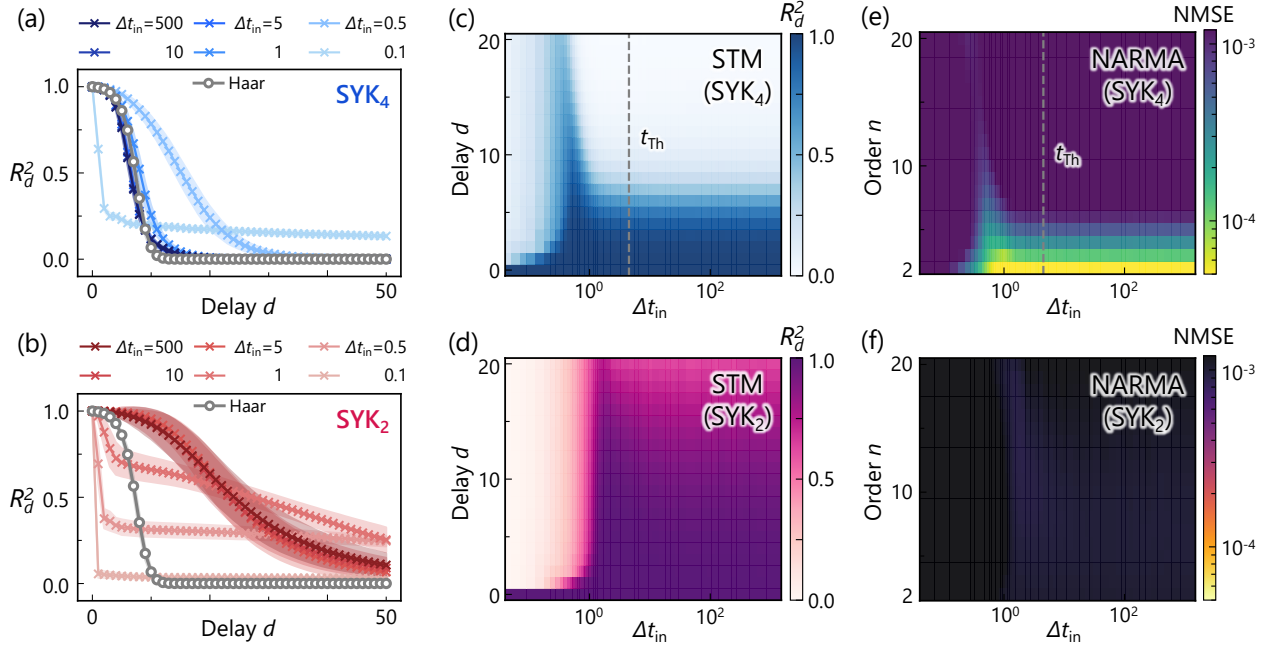


FIG. S8. (a), (b) QRC performance for the STM task R_d^2 as a function of delay d for various input intervals Δt_{in} , with shaded regions indicating standard deviations. The gray curve represents the Haar QRC performance. (c), (d) Heatmaps illustrating STM task performance, where the vertical axis represents the delay step d and the horizontal axis corresponds to the input interval Δt_{in} . (e), (f) Heatmaps of NARMA task performance. The vertical axis denotes the NARMA order n , and the horizontal axis shows the input interval Δt_{in} . Results for the SYK₄ model are presented in (a), (c), (e), while results for the SYK₂ model are in (b), (d), (f). The Thouless time t_{Th} is marked by vertical dashed lines in (c) and (e).

observed in localized phases to slow convergence. Our QRC likewise exhibits slow convergence in the SYK₂-dominant regime (Fig. S7). Nevertheless, as shown in Fig. 4(a) of the main text, the parametric edge of many-body quantum chaos appears at κ_2/J_4 values almost an order of magnitude smaller than those at which the distance metric first deviates from 0 in Fig. S7. Moreover, although the convergence metric varies significantly with κ_2/J_4 even for $\Delta t_{in} = 1$, the performance peak associated with the parametric edge does not appear in Fig. 4(b) of the main text. Taken together, these observations suggest that convergence and the parametric edge are not directly related here.

IV. RESERVOIR PERFORMANCE

The main text presents QRC performance on the short-term memory (STM) task and the nonlinear auto-regressive moving average (NARMA) task. Specifically, it benchmarks the SYK₄ [Fig. 3(c)] and SYK₂ [Fig. 3(d)] models in the temporal domain, as well as the interpolated model in the parametric domain at input intervals of $\Delta t_{in} = 50$ [Fig. 4(a)] and $\Delta t_{in} = 1$ [Fig. 4(b)]. The aim here is to provide additional data to further elucidate QRC performance and the edge of many-body quantum chaos. In Secs. IV A and IV B, we examine the STM and NARMA performance in the temporal and parametric domains, respectively, by sweeping the delay parameters

and task orders. We also report the QRC performance on additional tasks: the nonlinear STM task in Sec. IV C and the parity-check (PC) task in Sec. IV D. Finally, Sec. IV E investigates the effects of statistical noise on QRC performance.

A. Reservoir performance for the SYK₄ and SYK₂ models

This section examines the detailed QRC performance of the SYK₄ and SYK₂ models in the temporal domain. We first analyze performance on the STM task, which quantifies linear memory capability. Figures S8(a) and S8(b) present the memory performance R_d^2 as a function of delay d for each model, while the input interval Δt_{in} is varied. The performance of the Haar QRC model is included for baseline comparison. For the SYK₄ model, the performance R_d^2 approaches the Haar QRC reference value at large Δt_{in} , reflecting a universal RMT description when Δt_{in} exceeds the Thouless time t_{Th} . Optimal performance in this plot occurs at $\Delta t_{in} = 0.5$, before the Thouless time, where a long tail of nonzero R_d^2 extends to $d = 30$. Conversely, a very short input interval ($\Delta t_{in} = 0.1$) yields poor performance, with R_d^2 remaining around 0.2 even at large d . This degradation arises because the partial trace provides less effective memory erasure for such short input intervals, leading to the slow convergence observed in Fig. S7(a).

The tendency of short input intervals to introduce atypical decay in R_d^2 is more pronounced in the SYK₂ model

[Fig. S8(b)]. For instance, at $\Delta t_{\text{in}} \leq 1$, the memory performance drops abruptly instead of decaying gradually. This behavior suggests a compromised fading-memory property; due to the slow convergence, the system fails to sufficiently prioritize recent information. Nevertheless, for longer input intervals ($\Delta t_{\text{in}} = 5, 10$, and 500), QRC with the SYK₂ model demonstrates significantly better memory retention than the Haar QRC baseline, with a long tail spanning beyond $d = 50$. Importantly, this enhancement appears even though the convergence speed [quantified by $\|\rho_A^{(k)} - \rho_B^{(k)}\|$ in Fig. S6(b)] is comparable to that observed at shorter Δt_{in} . This discrepancy implies that, while the distance-based convergence metric is valuable for characterizing QRC systems, its direct connection to performance can be system-dependent. Specifically, in the SYK₂ model, ESP is satisfied only asymptotically, and it remains nontrivial to determine how this asymptotic nature precisely affects performance. Further investigation, potentially employing alternative convergence metrics, is left for future work.

Figures S8(c) and S8(d) extend Figs. S8(a) and S8(b), respectively, by presenting heatmaps of R_d^2 for $0 \leq d \leq 20$ as the input interval Δt_{in} is varied. In the SYK₄ model [Fig. S8(c)], a key observation is an oblique line of high performance that appears where Δt_{in} is shorter than the Thouless time. This sharp feature represents the performance enhancement at the temporal edge of many-body quantum chaos, consistent with the observation in Fig. 3(c) of the main text. Notably, as discussed there, the exact location of the performance peak is task-dependent. For the STM task investigated here, the peak shifts gradually to the nonchaotic side (smaller Δt_{in}) as d increases, leading to the observed oblique line. By contrast, a similar enhancement is absent in the SYK₂ model [Fig. S8(d)]; instead, it shows a binary separation between a regime of sufficient memory retention at large Δt_{in} and poor retention at small Δt_{in} , lacking any distinct performance peak.

Figures S8(e) and S8(f) show heatmaps of the NMSE for the NARMA tasks using the SYK₄ and SYK₂ models, respectively. The vertical axis shows the NARMA order n , and the horizontal axis indicates the input interval Δt_{in} . Consistent with the STM task results, a distinct performance peak is evident in the SYK₄ model, whereas the SYK₂ model shows only gradual performance changes. As discussed in the main text, the absence of performance peak in the SYK₂ model indicates that the performance enhancement observed in the SYK₄ model indeed originates from many-body quantum chaos. Notably, as the NARMA order n increases, the peak broadens and eventually vanishes, illustrating a regime in which the edge-induced peak vanishes due to excessive task complexity.

B. Reservoir performance for the interpolated model

Next, we analyze QRC performance of the model interpolating between the SYK₄ and SYK₂ systems in Fig. S9, extending the results presented in Fig. 4 of the main text.

Figures S9(a) and S9(b) illustrate the memory performance R_d^2 as a function of delay d for various coupling strengths κ_2 at $J_4 = 1$. For comparative purposes, the performance of the Haar QRC model is also plotted. In the long time-scale regime ($\Delta t_{\text{in}} = 50$), the QRC performance within the quantum chaotic regime ($\kappa_2 = 5.0$) aligns with the Haar QRC reference value, which is a direct consequence of universal RMT behavior [Fig. S9(a)]. As the coupling strength κ_2 increases, the STM performance improves monotonically, eventually exhibiting an extended tail in R_d^2 in the integrable regime ($\kappa_2 = 100.0$). This trend is further elucidated in the corresponding heatmap in Fig. S9(c), which plots R_d^2 against the coupling strength ratio κ_2/J_4 (horizontal axis) and delay d (vertical axis). The onsets of the chaotic and integrable regimes are indicated by gray dashed lines. Within the quantum chaotic regime, the performance is nearly uniform across κ_2/J_4 , showing a similar dependence on delay d . Upon entering the intermediate regime, R_d^2 gradually improves towards the integrable regime, where memory retention is strongest. The boundary separating high (red) and low (blue) R_d^2 values forms an oblique line, indicating that stronger κ_2 coupling is necessary to preserve memory for longer delays d .

By contrast, in the short time-scale regime ($\Delta t_{\text{in}} = 1$), R_d^2 displays a markedly different pattern: a sharp initial drop followed by a slow decay, the decay rate of which diminishes further as the system approaches the integrable limit at large κ_2 [Fig. S9(b)]. The heatmap in Fig. S9(d) corroborates this observation: a region of high R_d^2 (red) is concentrated at small delays (bottom), illustrating the initial high performance before an abrupt drop. The subsequent slow decay corresponds to the upper regions of nearly uniform, low R_d^2 (blue). These behaviors are attributed to the less effective memory erasure via the partial trace operation when the input interval Δt_{in} is short, as discussed previously.

We additionally present performance heatmaps for the NARMA task in Fig. S9(e) for $\Delta t_{\text{in}} = 50$ and Fig. S9(f) for $\Delta t_{\text{in}} = 1$. The horizontal axis represents the coupling strength κ_2/J_4 , and the vertical axis shows the NARMA order n . For small NARMA orders with a long input interval [Fig. S9(e)], the performance is distinctly enhanced around the chaotic boundary, corresponding to the parametric edge of many-body quantum chaos. This edge-induced performance enhancement is absent for the short input interval $\Delta t_{\text{in}} = 1$ in Fig. S9(f) because, as discussed in the main text, the QRC requires a sufficiently long input interval to manifest its quantum chaotic characteristics. As the NARMA order n increases, the task becomes substantially more challenging, and the QRC can no longer represent it adequately, as evidenced by the degraded performance. In this regime with large n , the effective strategy is essentially limited to prioritizing the representation of contributions driven by distant past inputs, rather than capturing complex nonlinear details. Accordingly, in the high- n regime the performance contours for the NARMA task [Fig. S9(e)] resemble those for the STM task [Fig. S9(c)].

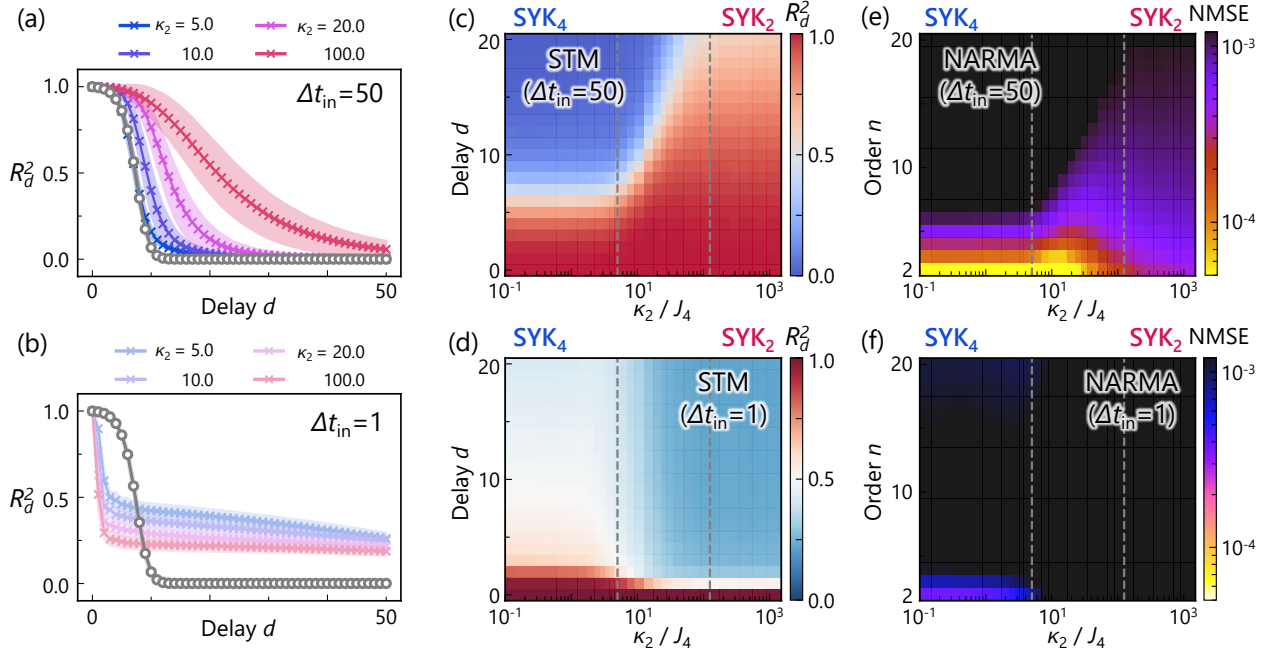


FIG. S9. (a), (b) QRC performance on the STM task R_d^2 as a function of delay d . Different colored lines represent various coupling strengths κ_2 , with shaded areas denoting standard deviations. The gray curve represents the Haar QRC performance. (c), (d) Heatmap representation of the STM task performance R_d^2 . The vertical axis represents the delay step d , and the horizontal axis corresponds to the ratio of coupling strength κ_2 / J_4 . (e), (f) Heatmaps of NARMA task performance. The vertical axis denotes the NARMA order n , while the horizontal axis shows the coupling strength κ_2 / J_4 . In heatmaps (c)-(f), vertical dashed lines mark the approximate onset of WD (left) and Poisson (right) statistical distributions in the spectral properties. Panels (a), (c), (e) correspond to the input interval of $\Delta t_{\text{in}} = 50$, while panels (b), (d), (f) correspond to $\Delta t_{\text{in}} = 1$.

C. Nonlinear STM task

To further benchmark the QRC, we employ the nonlinear extension of the STM task [14, 15]. This task is designed to systematically vary the relative demands on memory and nonlinear processing. As in the standard STM task, a sequence of random inputs $\{u^{(k)}\}$, with $u^{(k)} \in [0, 1]$, is provided to the quantum reservoir. For a given polynomial order n and delay d , the target output at time step k is defined as $\bar{y}^{(k)} = (u^{(k-d)})^n$. The case $n = 1$ recovers the standard (linear) STM task analyzed in the main text. By increasing n at fixed d , the task selectively enhances the demand for nonlinear processing, while the memory requirement (determined by d) remains unchanged. The QRC performance on this task is quantified using the determination coefficient R_d^2 .

We present the performance on the nonlinear STM task with delay $d = 4$ around the temporal and parametric edges of many-body quantum chaos in Figs. S10(a) and S10(b), respectively. In particular, the temporal edge is studied using the SYK₄ model, while the parametric edge is examined using the interpolated model at $\Delta t_{\text{in}} = 50$. In both cases, the performance decreases monotonically with increasing the nonlinearity order n , reflecting the greater task complexity. Notably, in the temporal domain, the distinct performance peak at the temporal edge persists from low- to high-order nonlinearities.

The position of this peak appears to be independent of the nonlinearity order n , suggesting that the location of the temporal edge for this task is governed by the memory properties.

In the parametric domain, the performance peak is blurred for the linear STM task ($n = 1$), because memory retention is strongest in the integrable regime. However, as the nonlinearity is introduced to the task via $n \geq 2$, QRC performance in the integrable regime degrades more rapidly than in the chaotic regime, reflecting the former's limited nonlinear transformation capability. Consequently, a performance peak at the parametric edge of many-body quantum chaos becomes prominent for these tasks. Remarkably, with increasing n , the peak position shift towards smaller κ_2 / J_4 values, i.e., closer to the chaotic regime. This shift reflects that operating closer to the chaotic side enhances the nonlinear transformation capacity, which is required to solve tasks with higher nonlinearity order n .

D. PC task

We next evaluate the PC task [15–18]. In this task, binary inputs $u'^{(k)} \in \{0, 1\}$ are injected into the quantum reservoir, and the target output is the parity of the sum of the previous inputs: $\bar{y}^{(k)} = (\sum_{d=0}^n u'^{(k-d)}) \bmod 2$. We also use the determi-

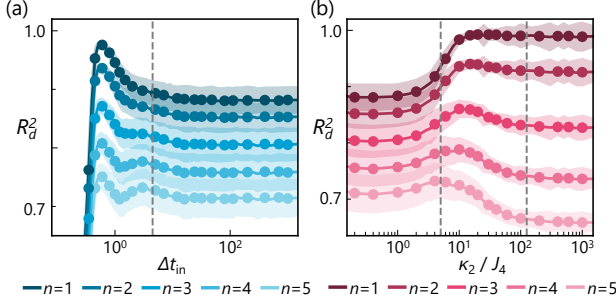


FIG. S10. QRC performance on the nonlinear STM task with delay $d = 4$ for orders $n = 1$ to 5 . (a) Performance of the SYK₄ model plotted as a function of the input interval Δt_{in} . The Thouless time t_{th} is indicated by the vertical dashed line. (b) Performance of the interpolated model as a function of the coupling strength κ_2/J_4 , with the input interval fixed at $\Delta t_{\text{in}} = 50$. The onsets of the chaotic and integrable regimes are marked by the left and right horizontal lines, respectively.

nation coefficient R^2 to evaluate performance on this task. A defining feature of this task is its strong nonlinearity requirement. The PC task of order n requires memory spanning $n + 1$ steps (the current input plus the n preceding ones). More importantly, it requires nonlinearity of order $n + 1$ involving many cross-term interactions among delayed inputs, because the parity function corresponds to an $(n + 1)$ -degree polynomial in its inputs. In other words, increasing n by one adds only a single additional step to the memory requirement, whereas the nonlinearity requirement grows much more sharply both in polynomial order and in the combinatorial complexity of cross-term interactions.

Figures S11(a) and S11(b) presents the QRC performance for the PC task with $n = 1, 2$, and 3 , evaluated in the temporal domain [Fig. S11(a)] and the parametric domain [Fig. S11(b)]. For $n = 1$, we observe clear performance peaks at the edge of many-body quantum chaos in both cases. In contrast, for $n = 2$ and 3 , the task's nonlinearity requirement escalates sharply and overwhelms the memory demand. Consequently, optimal performance is achieved within the chaotic regime, where the nonlinearity of the input-to-output mapping is strongest. This behavior exemplifies an extreme imbalance in task requirements: when either memory or nonlinear transformation dominates, the optimal operating regime is integrable or chaotic, respectively, and the edge-induced performance peak is absent. The linear STM task (memory-dominated) and the PC task with $n \geq 2$ (nonlinearity-dominated) exemplify these extremes. It should be noted that in such limits, explicit design guidelines are less necessary because the optimal operating regime follows directly from the task definition.

Having observed that the edge-induced peak appears when a nonlinearity requirement is added to an otherwise memory-dominant STM task (the nonlinear STM task in Fig. S10), we next examine whether introducing an additional memory requirement to the PC task likewise produces such a

peak. To this end, we modify the target output to $\bar{y}^{(k)} = \left(\sum_{d=0}^n u^{(k-d-b)} \right) \bmod 2$, where b denotes the temporal bias step. Figures S11(c) and S11(d) present the QRC performance for the PC task $n = 1, 2, 3$ under the bias $b = 2$. Strikingly, a clear performance enhancement at the edge of many-body quantum chaos is observed for both edges, across all orders investigated. Because the bias b increases only the memory requirement while leaving the nonlinearity unchanged, these results support the interpretation that the absence of a performance peak in the original PC task is due to the dominance of the nonlinear requirement. Introducing the bias makes memory a performance limiting factor alongside nonlinearity, thereby resolving the imbalance in task demands. The resulting setting thus becomes analogous to the NARMA task and the nonlinear STM task, for which both memory and nonlinearity constrain performance; in such nontrivial regimes, the edge of many-body quantum chaos becomes most prominent. Although not known a priori, realistic scenarios involving real-world data are expected to fall into this category. The pronounced performance peaks observed for these tasks therefore highlight the importance of operating at the edge of many-body quantum chaos as a robust design guideline for QRC. Finally, we note that increasing the order n in the biased PC task shifts the peak toward the chaotic regime, paralleling the trend observed in the nonlinear STM task (Fig. S10).

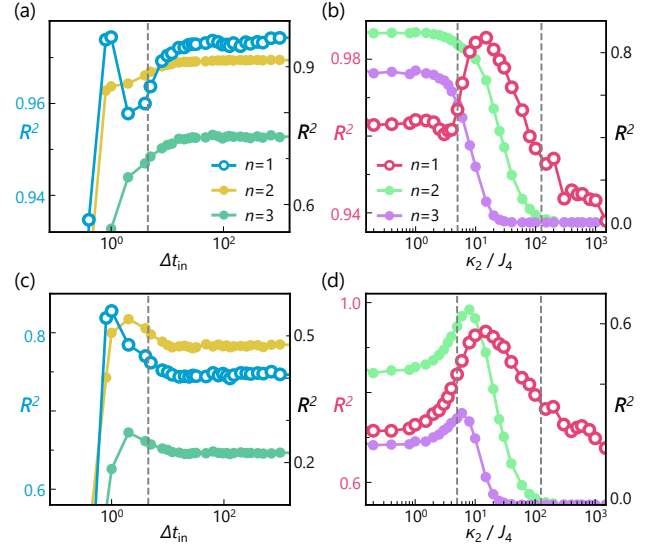


FIG. S11. QRC performance on (a), (b) the PC task and (c), (d) the PC task with memory bias $b = 2$. (a), (c) Performance using the SYK₄ model as a function of Δt_{in} . (b), (d) Performance using the interpolated model as a function of κ_2/J_4 , with $\Delta t_{\text{in}} = 50$. In all panels, R^2 values for $n = 1$ are plotted on the left vertical axis, and values for $n = 2$ and 3 are on the right vertical axis.

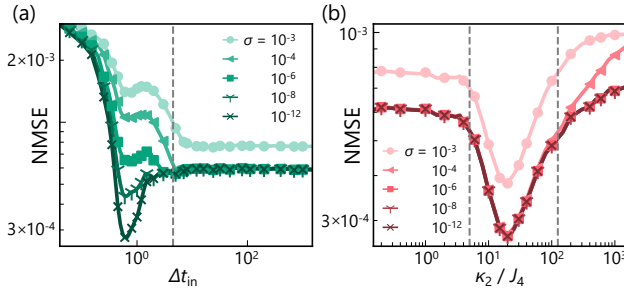


FIG. S12. The QRC performance (NMSE) on the NARMA-5 task under sampling noise. (a) The NMSE for the SYK₄ model as a function of the input interval Δt_{in} . (b) The NMSE for the interpolated model as a function of the coupling strength κ_2/J_4 , with Δt_{in} fixed at 50. In both panels, the colors and markers indicate the sampling noise strength σ .

E. Impact of statistical uncertainty in measurements

Because expectation values are estimated from a finite number of measurement shots, their accuracy is limited by sampling noise. Our analysis thus far has assumed ideal, noise-free conditions; we now investigate the impact of statistical noise on QRC performance by modeling finite sampling error as additive Gaussian noise. Specifically, to the noiseless expectation value $n_i = \langle c_i^\dagger c_i \rangle$, we add a zero-mean Gaussian noise with standard deviation $\sigma_i = \sigma \sqrt{n_i(1-n_i)}$, where σ parametrizes the shot noise strength [19–21]. This variance matches the binomial variance of repeated projective measurements in the $c_i^\dagger c_i$ basis, and the Gaussian approximation is valid when $\sigma \ll 1$.

In Figs. S12(a) and S12(b), we show the NARMA-5 performance around the temporal edge and parametric edge, respectively, as the sampling noise strength σ is varied. As expected for algorithms that rely on estimating expectation values, statistical fluctuations degrade the performance; accordingly, reducing σ monotonically improves it. Note that the sensitivity to this noise depends on the operating regime. In the temporal sweep [Fig. S12(a)], small input intervals Δt_{in} are markedly more vulnerable than large Δt_{in} (the RMT regime), because the system-specific structure is more pronounced at small Δt_{in} and is therefore more susceptible to statistical noise. For large σ , this degradation overwhelms the edge-induced enhancement, producing local minima near the temporal edge and shifting the global minimum to the large- Δt_{in} regime. As σ is reduced to 10^{-8} , the temporal edge again yields the global minimum; at $\sigma = 10^{-12}$, it nearly recovers the ideal performance shown in Fig. 3(c) of the main text. By contrast, in the parametric sweep we fix $\Delta t_{\text{in}} = 50$, where the QRC is comparatively robust to sampling noise. Consequently, the parametric edge of many-body quantum chaos attains the global minimum even at $\sigma = 10^{-3}$, and the performance is essentially converged to the ideal from $\sigma = 10^{-4}$ onward.

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